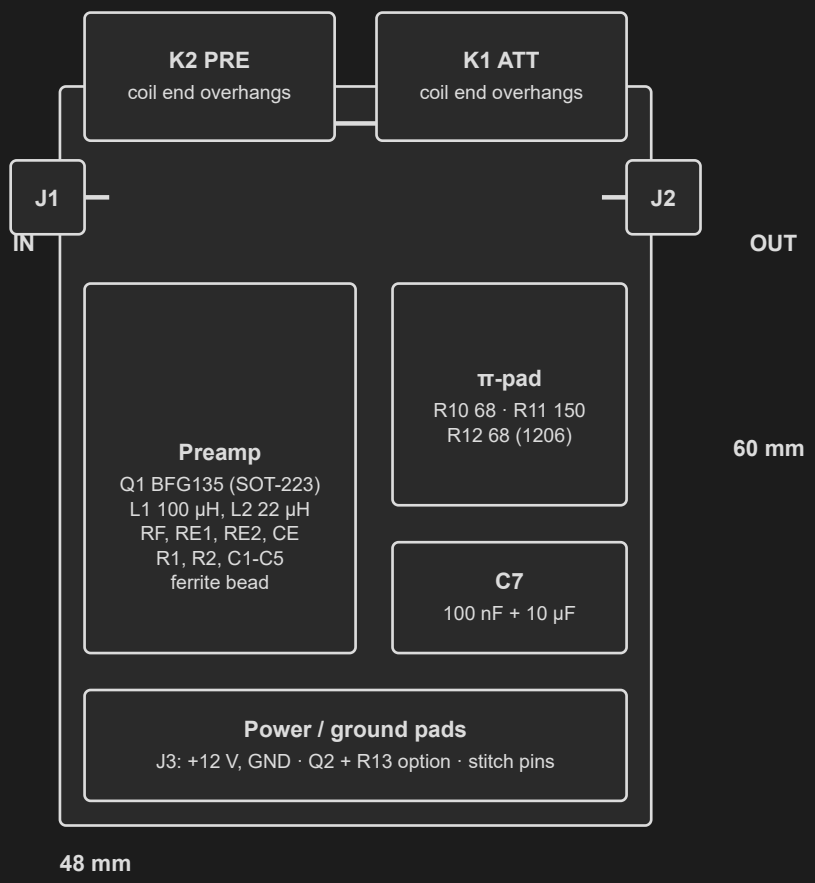
ATT / PRE board — PCB plan for EasyEDA

48 × 60 mm, on the crystal filter can. IN top left → K2 PRE → K1 ATT → OUT top right. Preamp on the left (away from the Nano), pad on the right.

Floor plan, to scale (6 px = 1 mm), component side



Net list

Net	Connects
RF_IN	J1 signal, K2 COM-A
PRE_BYP	K2 NC-A, K2 NC-B (short wide track)
PRE_IN	K2 NO-A, C1
BASE	C1, Q1 B, L2, RF-feedback (RF), (bias via L2)
BIAS	L2, R1, R2, C5
EMIT	Q1 E, RE1
E2	RE1, RE2, CE
COLL	Q1 C (tab), ferrite bead
COLL2	ferrite bead, L1, C4, C3
FB	C3, RF
PRE_OUT	C4, K2 NO-B
MID	K2 COM-B, K1 COM-A
ATT_BYP	K1 NC-A, K1 NC-B (short wide track)
PAD_IN	K1 NO-A, R10, R11
PAD_OUT	R11, R12, K1 NO-B
RF_OUT	K1 COM-B, J2 signal
V12	J3-1, C7, R6 (Q2 emitter if fitted)
V12F	R6, C6, R1, L1 (Q2 collector feeds R6 if fitted)
GND	J1/J2 shields, J3-2, R2, R10, R12, RE2, CE, C5, C6, C7, bottom plane

Relay pin names are generic (COM / NC / NO, poles A and B): map them to the Finder 30.22 footprint (NC = rest position). K2 = PRE, K1 = ATT. Values on the ATT / PRE board sheet.

Design rules (home etching)

- Parts 1206 (resistors, capacitors), BFG135 SOT-223, chokes 1210 or axial.
- RF tracks **1.5 mm**, power **2 mm**, others ≥ 1 mm; clearance ≥ **0.6 mm**.
- Top: tracks + ground pour (0.8 mm clearance). Bottom: **unetched solid copper = ground plane**.
- Top ground to bottom: wire pins through drilled holes, at least next to every GND pad of the RF parts, at J1 / J2 and around the preamp.
- Relay holes: no relay pin is GND — clear the bottom copper (ø 2.5 mm) around **every** relay pin and every other non-GND through-hole.

- Coil pins overhang the edge: flyback 1N4148 and sense diode soldered directly on the coil pins, wires to the rocker / Nano.